

1(a)	A – line to epidermal cell ; B – line to mesophyll cell ;	2
1(b)(i)	<u>bundle sheath</u> (cell) ;	1
1(b)(ii)	any four from: 1 decrease / stop, <u>photorespiration</u> ; 2 <u>mesophyll</u> cells, surround / protect / form ring round, bundle sheath ; 3 air / oxygen, cannot reach, bundle sheath (cells) ; 4 oxygen cannot, reach / bind to, RuBP / rubisco (in BSC) ; 5 high / much, carbon dioxide (in bundle sheath cells) ; 6 AVP ;	4
1(c)(i)	any four from: 1 as temperature increases rate increases, up to 25 °C / at low temperatures, for both ; 2 as temperature increases rate decreases, above 35 °C / at high temperatures, for both ; 3 C3 rate <u>higher</u>, at low temperatures / < 22 °C ; 4 C3 rate <u>lower</u>, at high temperatures / > 22 °C ; 5 C3, peak rate / maximum rate / optimum T °C, at 25 °C and C4, peak rate / maximum rate / optimum T °C, at 30–35 °C ; 6 C3 has <u>lower</u>, maximum rate / peak ; 7 C3, maximum rate / peak, is at <u>lower</u> temperature or C3 has a lower optimum temperature ; 8 comparative data quote ;	4
1(c)(ii)	<i>comparison</i> rate, <u>higher</u> / <u>faster</u>, high CO₂ conditions ; (at 30–35 °C) rate is maximum for both ; <i>explanation (max 1)</i> carbon dioxide, limits rate / is limiting factor, in low CO₂ conditions ; AVP ;	2

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